POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6043B | Semester VI | Theory | 4 Credits

Medical Electronics

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6043B as Medical Electronics in Semester VI for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Madras’s Biomedical Instrumentation course and polytechnic medical electronics curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Medical designs, bio-signal acquisitions, therapeutic protocols and imaging systems must be based on approved engineering plans, certified medical data, current safety standards and competent professional review.

Verified source note: Verified sources support bio-potentials, medical sensors, therapeutic devices, and patient safety. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Medical Electronics studies the application of electronic principles and devices to the field of medicine for diagnosis, monitoring and treatment. It develops the ability to analyze bio-potentials (ECG, EEG), evaluate medical sensors and imaging systems, understand therapeutic equipment like pacemakers, and coordinate patient-safe electronic solutions while respecting the technical and clinical boundaries of electronic and biomedical engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Basic electronics, operational amplifiers, digital electronics and human anatomy basics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the origin of bio-potentials and the characteristics of various medical sensors.
2. Analyze the acquisition and amplification of physiological signals like ECG and EEG.
3. Evaluate the operation and applications of diagnostic and therapeutic medical equipment.
4. Explain the systematic safety standards, imaging systems and future trends in medical electronics.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes		
CO	**Official outcome**	**Bloom level**
CO1	Explain the origin of bio-potentials and characteristics of medical sensors.	Understand
CO2	Analyze the acquisition and processing of bio-signals like ECG and EEG.	Apply
CO3	Evaluate the operation of therapeutic and diagnostic medical equipment.	Apply
CO4	Explain medical imaging systems and patient safety standards.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Bio-potentials and Medical Sensors	Origin of bio-potentials: Resting and Action potential; Sodium-Potassium pump; Medical sensors: Electrodes (Surface, Needle), Transducers for pressure, temperature and flow.	Approx. 14 hours	CO1	Module 1
2	Bio-signal Acquisition and Amplifiers	ECG, EEG, EMG: Lead systems and waveforms; Bio-amplifiers: Differential, Instrumentation, and Isolation amplifiers; Interference rejection; Signal conditioning basics.	Approx. 16 hours	CO2	Module 2
3	Diagnostic and Therapeutic Equipment	Patient monitoring systems; Therapeutic devices: Cardiac pacemakers (Internal/External), Defibrillators, Ventilators, Surgical diathermy (Electrosurgery).	Approx. 16 hours	CO3	Module 3
4	Medical Imaging and Patient Safety	Medical imaging: X-ray, CT, MRI, Ultrasound basics; Patient safety: Macroshock, Microshock, Leakage current; Safety standards (IEC 60601); Future trends.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Bio-potentials and Medical Sensors

Origin of bio-potentials: Resting and Action potential; Sodium-Potassium pump; Medical sensors: Electrodes (Surface, Needle), Transducers for pressure, temperature and flow.

CO1 · Approx. 14 hours

Bio-signal Acquisition and Amplifiers

ECG, EEG, EMG: Lead systems and waveforms; Bio-amplifiers: Differential, Instrumentation, and Isolation amplifiers; Interference rejection; Signal conditioning basics.

CO2 · Approx. 16 hours

Diagnostic and Therapeutic Equipment

Patient monitoring systems; Therapeutic devices: Cardiac pacemakers (Internal/External), Defibrillators, Ventilators, Surgical diathermy (Electrosurgery).

CO3 · Approx. 16 hours

Medical Imaging and Patient Safety

Medical imaging: X-ray, CT, MRI, Ultrasound basics; Patient safety: Macroshock, Microshock, Leakage current; Safety standards (IEC 60601); Future trends.

CO4 · Approx. 14 hours

MODULE 1

Bio-potentials and Medical Sensors

Origin of bio-potentials: Resting and Action potential; Sodium-Potassium pump; Medical sensors: Electrodes (Surface, Needle), Transducers for pressure, temperature and flow.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

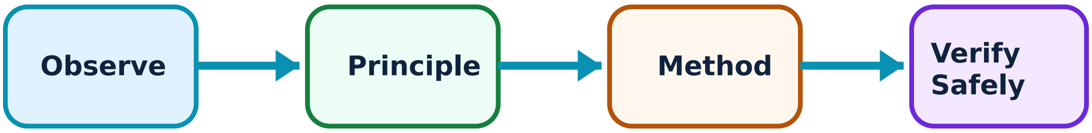
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Bio-potentials and Medical Sensors: identify the system, state its principle, follow the correct method and verify the result safely.

1. Bio-potential Fundamentals−

Bio-potentials are electrical signals generated by biological cells. The resting potential is maintained by ion pumps. Stimulation creates an action potential. These signals are the basis for diagnostic tools like ECG (heart), EEG (brain), and EMG (muscle).

Study and application method

Describe the 'Action Potential' generation conceptually; explain the role of 'Ion Channels' in bio-potential formation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the 'Action Potential' generation conceptually; explain the role of 'Ion Channels' in bio-potential formation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Bio-potentials are extremely weak and prone to noise. Do not use standard electronic probes for bio-signal acquisition without authorised high-impedance, low-noise medical electrodes.

2. Medical Electrodes and Transducers−

Electrodes interface the body with the instrument. Surface electrodes are non-invasive. Transducers convert physiological parameters like blood pressure (piezoelectric) and body temperature (thermistors) into electrical signals for monitoring.

Study and application method

Compare 'Surface' and 'Needle' electrodes conceptually; describe the working of a 'Thermistor' for medical temperature measurement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Medical sensors must be biocompatible. Do not use sensors for internal measurements without authorised sterilization and biocompatibility certification.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Bio-signal Acquisition and Amplifiers

ECG, EEG, EMG: Lead systems and waveforms; Bio-amplifiers: Differential, Instrumentation, and Isolation amplifiers; Interference rejection; Signal conditioning basics.

Approx. 16 hours **CO2** **Understand · Apply**

Learning goals

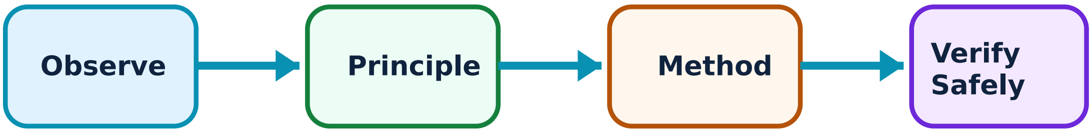
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Bio-signal Acquisition and Amplifiers: identify the system, state its principle, follow the correct method and verify the result safely.

1. Bio-signal Characteristics−

ECG records heart electrical activity via specific lead systems (e.g., Einthoven's triangle). EEG records brain waves (Alpha, Beta, Delta). Each signal has unique frequency and amplitude characteristics that dictate the design of the acquisition system.

Study and application method

Sketch a typical 'ECG Waveform' and label its components; list the frequency range of 'EEG' waves.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a typical 'ECG Waveform' and label its components; list the frequency range of 'EEG' waves.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Power line interference (50/60Hz) is a major issue. All bio-signal acquisition must include authorised notch filters and right-leg drive (RLD) circuits for interference rejection.

2. Medical Amplifiers−

Bio-amplifiers require high input impedance and high CMRR (Common Mode Rejection Ratio). Isolation amplifiers are critical for patient safety, providing electrical isolation between the patient and the mains-powered equipment.

Study and application method

Explain the importance of 'Isolation' in medical amplifiers; describe the role of an 'Instrumentation Amplifier' in bio-signal gain.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the importance of 'Isolation' in medical amplifiers; describe the role of an 'Instrumentation Amplifier' in bio-signal gain.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Amplifier leakage current can be fatal. Do not use bio-amplifiers without authorised verification of patient leakage current limits as per safety standards.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Diagnostic and Therapeutic Equipment

Patient monitoring systems; Therapeutic devices: Cardiac pacemakers (Internal/External), Defibrillators, Ventilators, Surgical diathermy (Electrosurgery).

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

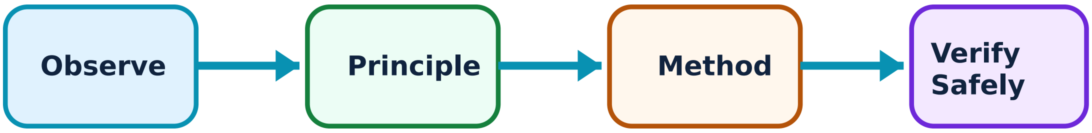
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Diagnostic and Therapeutic Equipment: identify the system, state its principle, follow the correct method and verify the result safely.

1. Patient Monitoring−

Bedside monitors track vital signs like heart rate, respiration, and SpO2. Alarms are used to alert staff to critical changes. Modern systems use digital processing and networking for remote monitoring in ICUs.

Study and application method

List the parameters monitored in a standard 'Bedside Monitor'; explain the principle of 'Pulse Oximetry' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the parameters monitored in a standard 'Bedside Monitor'; explain the principle of 'Pulse Oximetry' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Alarm fatigue can lead to missed events. All monitoring systems must follow authorised alarm-management protocols and sensor-integrity checks.

2. Therapeutic Devices−

Pacemakers maintain heart rhythm. Defibrillators restore rhythm during cardiac arrest. Ventilators assist breathing. Electrosurgery (diathermy) uses high-frequency current for cutting and coagulation during surgery.

Study and application method

Differentiate between 'Internal' and 'External' pacemakers; describe the operation of a 'DC Defibrillator' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Internal' and 'External' pacemakers; describe the operation of a 'DC Defibrillator' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Electrosurgery involves high-frequency electrical hazards. Do not operate diathermy equipment without authorised training and adherence to grounding and return-electrode safety.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Medical Imaging and Patient Safety

Medical imaging: X-ray, CT, MRI, Ultrasound basics; Patient safety: Macroshock, Microshock, Leakage current; Safety standards (IEC 60601); Future trends.

Approx. 14 hours | **CO4** | **Understand · Apply**

Learning goals

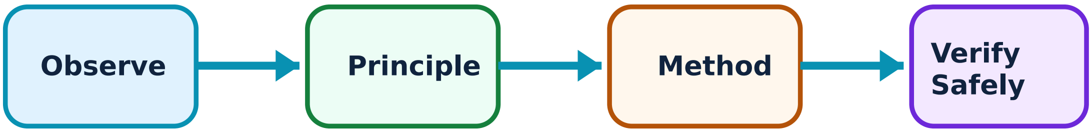
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Medical Imaging and Patient Safety: identify the system, state its principle, follow the correct method and verify the result safely.

1. Medical Imaging Systems−

Imaging allows non-invasive visualization. X-rays and CT use radiation. MRI uses magnetic fields. Ultrasound uses sound waves. Each has specific applications and safety requirements.

Study and application method

Compare 'X-ray' and 'Ultrasound' imaging conceptually; explain the advantage of 'MRI' for soft tissue visualization.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'X-ray' and 'Ultrasound' imaging conceptually; explain the advantage of 'MRI' for soft tissue visualization.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: X-rays involve ionizing radiation. All imaging procedures must follow authorised radiation protection guidelines (ALARA principle).

2. Patient Safety and Standards−

Safety is paramount. Macroshock occurs through the skin, while microshock is direct to the heart. IEC 60601 is the global safety standard. Regular testing for leakage current and grounding is mandatory for all medical devices.

Study and application method

Define 'Microshock' and explain its danger; list three electrical safety measures used in hospitals.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Microshock' and explain its danger; list three electrical safety measures used in hospitals.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Medical equipment must be regularly calibrated and safety-tested. Do not use medical devices without authorised periodic safety audits as per IEC 60601 standards.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Bio-potentials and Medical Sensors

Use the concept flow to revise observation, principle, method and verification.

Module 2: Bio-signal Acquisition and Amplifiers

Use the concept flow to revise observation, principle, method and verification.

Module 3: Diagnostic and Therapeutic Equipment

Use the concept flow to revise observation, principle, method and verification.

Module 4: Medical Imaging and Patient Safety

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare ECG and EEG waveforms in a conceptual table showing amplitude and frequency differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the electrode placement for 'Einthoven's Triangle' conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the block diagram of an Isolation Amplifier showing the isolation barrier.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the CMRR conceptually for a given differential amplifier stage.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a medical electronic device (e.g., BP monitor, Thermometer) and note its electronic features.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Bio-potentials and Medical Sensors

Explain Bio-potential Fundamentals and state one correct method, application or precaution.−

Bio-potentials are electrical signals generated by biological cells. The resting potential is maintained by ion pumps. Stimulation creates an action potential. These signals are the basis for diagnostic tools like ECG (heart), EEG (brain), and EMG (muscle).

Answer framework: Describe the 'Action Potential' generation conceptually; explain the role of 'Ion Channels' in bio-potential formation.

Important point: Bio-potentials are extremely weak and prone to noise. Do not use standard electronic probes for bio-signal acquisition without authorised high-impedance, low-noise medical electrodes.

Explain Medical Electrodes and Transducers and state one correct method, application or precaution.−

Electrodes interface the body with the instrument. Surface electrodes are non-invasive. Transducers convert physiological parameters like blood pressure (piezoelectric) and body temperature (thermistors) into electrical signals for monitoring.

Answer framework: Compare 'Surface' and 'Needle' electrodes conceptually; describe the working of a 'Thermistor' for medical temperature measurement.

Important point: Medical sensors must be biocompatible. Do not use sensors for internal measurements without authorised sterilization and biocompatibility certification.

Module 2 — Bio-signal Acquisition and Amplifiers

Explain Bio-signal Characteristics and state one correct method, application or precaution.−

ECG records heart electrical activity via specific lead systems (e.g., Einthoven's triangle). EEG records brain waves (Alpha, Beta, Delta). Each signal has unique frequency and amplitude characteristics that dictate the design of the acquisition system.

Answer framework: Sketch a typical 'ECG Waveform' and label its components; list the frequency range of 'EEG' waves.

Important point: Power line interference (50/60Hz) is a major issue. All bio-signal acquisition must include authorised notch filters and right-leg drive (RLD) circuits for interference rejection.

Explain Medical Amplifiers and state one correct method, application or precaution.−

Bio-amplifiers require high input impedance and high CMRR (Common Mode Rejection Ratio). Isolation amplifiers are critical for patient safety, providing electrical isolation between the patient and the mains-powered equipment.

Answer framework: Explain the importance of 'Isolation' in medical amplifiers; describe the role of an 'Instrumentation Amplifier' in bio-signal gain.

Important point: Amplifier leakage current can be fatal. Do not use bio-amplifiers without authorised verification of patient leakage current limits as per safety standards.

Module 3 — Diagnostic and Therapeutic Equipment

Explain Patient Monitoring and state one correct method, application or precaution.−

Bedside monitors track vital signs like heart rate, respiration, and SpO2. Alarms are used to alert staff to critical changes. Modern systems use digital processing and networking for remote monitoring in ICUs.

Answer framework: List the parameters monitored in a standard 'Bedside Monitor'; explain the principle of 'Pulse Oximetry' conceptually.

Important point: Alarm fatigue can lead to missed events. All monitoring systems must follow authorised alarm-management protocols and sensor-integrity checks.

Explain Therapeutic Devices and state one correct method, application or precaution. −

Pacemakers maintain heart rhythm. Defibrillators restore rhythm during cardiac arrest. Ventilators assist breathing. Electrosurgery (diathermy) uses high-frequency current for cutting and coagulation during surgery.

Answer framework: Differentiate between 'Internal' and 'External' pacemakers; describe the operation of a 'DC Defibrillator' conceptually.

Important point: Electrosurgery involves high-frequency electrical hazards. Do not operate diathermy equipment without authorised training and adherence to grounding and return-electrode safety.

Module 4 — Medical Imaging and Patient Safety

Explain Medical Imaging Systems and state one correct method, application or precaution.−

Imaging allows non-invasive visualization. X-rays and CT use radiation. MRI uses magnetic fields. Ultrasound uses sound waves. Each has specific applications and safety requirements.

Answer framework: Compare 'X-ray' and 'Ultrasound' imaging conceptually; explain the advantage of 'MRI' for soft tissue visualization.

Important point: X-rays involve ionizing radiation. All imaging procedures must follow authorised radiation protection guidelines (ALARA principle).

Explain Patient Safety and Standards and state one correct method, application or precaution.−

Safety is paramount. Macroshock occurs through the skin, while microshock is direct to the heart. IEC 60601 is the global safety standard. Regular testing for leakage current and grounding is mandatory for all medical devices.

Answer framework: Define 'Microshock' and explain its danger; list three electrical safety measures used in hospitals.

Important point: Medical equipment must be regularly calibrated and safety-tested. Do not use medical devices without authorised periodic safety audits as per IEC 60601 standards.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Bio-potentials and Medical Sensors.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Bio-signal Acquisition and Amplifiers.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Diagnostic and Therapeutic Equipment.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Medical Imaging and Patient Safety.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.

2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Origin of bio-potentials: Resting and Action potential; Sodium-Potassium pump; Medical sensors: Electrodes (Surface, Needle), Transducers for pressure, temperature and flow..
2. Develop a complete answer or practical plan for: ECG, EEG, EMG: Lead systems and waveforms; Bio-amplifiers: Differential, Instrumentation, and Isolation amplifiers; Interference rejection; Signal conditioning basics..
3. Develop a complete answer or practical plan for: Patient monitoring systems; Therapeutic devices: Cardiac pacemakers (Internal/External), Defibrillators, Ventilators, Surgical diathermy (Electrosurgery)..
4. Develop a complete answer or practical plan for: Medical imaging: X-ray, CT, MRI, Ultrasound basics; Patient safety: Macroshock, Microshock, Leakage current; Safety standards (IEC 60601); Future trends..

LAST-MILE REVISION

Quick Revision

Module 1: Bio-potentials and Medical Sensors

Origin of bio-potentials: Resting and Action potential; Sodium-Potassium pump; Medical sensors: Electrodes (Surface, Needle), Transducers for pressure, temperature and flow.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Bio-signal Acquisition and Amplifiers

ECG, EEG, EMG: Lead systems and waveforms; Bio-amplifiers: Differential, Instrumentation, and Isolation amplifiers; Interference rejection; Signal conditioning basics.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Diagnostic and Therapeutic Equipment

Patient monitoring systems; Therapeutic devices: Cardiac pacemakers (Internal/External), Defibrillators, Ventilators, Surgical diathermy (Electrosurgery).

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Medical Imaging and Patient Safety

Medical imaging: X-ray, CT, MRI, Ultrasound basics; Patient safety: Macroshock, Microshock, Leakage current; Safety standards (IEC 60601); Future trends.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Bio-potential Fundamentals	Bio-potentials are electrical signals generated by biological cells.
Medical Electrodes and Transducers	Electrodes interface the body with the instrument.
Bio-signal Characteristics	ECG records heart electrical activity via specific lead systems (e.
Medical Amplifiers	Bio-amplifiers require high input impedance and high CMRR (Common Mode Rejection Ratio).
Patient Monitoring	Bedside monitors track vital signs like heart rate, respiration, and SpO2.
Therapeutic Devices	Pacemakers maintain heart rhythm.
Medical Imaging Systems	Imaging allows non-invasive visualization.
Patient Safety and Standards	Safety is paramount.

LEARNING RESOURCES

References

Recommended medical electronics references

1. L. Cromwell, F. J. Weibell and E. A. Pfeiffer, Biomedical Instrumentation and Measurements.
2. R. S. Khandpur, Handbook of Biomedical Instrumentation.
3. John G. Webster, Medical Instrumentation: Application and Design.
4. Joseph J. Carr and John M. Brown, Introduction to Biomedical Equipment Technology.

Approved public medical electronics sources

- https://nptel.ac.in/courses/102106669
- https://onlinecourses.nptel.ac.in/noc25_bt76/preview

These sources provide the verified study basis while official Course 6043B detail is unavailable; they do not replace official clinical designs, authorised medical device maintenance, certified equipment specifications or authorised hospital safety protocols.